Advanced Representation Learning
(고급표현학습)

Instructor: Donghyeon Cho

The Department of Computer Science

Hanyang University

한양대학교
HANYANG UNIVERSITY

Contents of Table

- 1. Weakly-Supervised Learning
- 2. Challenge
- 3. Weakly-Supervised Object Detection
- 4. Deep and End-to-End WSOD

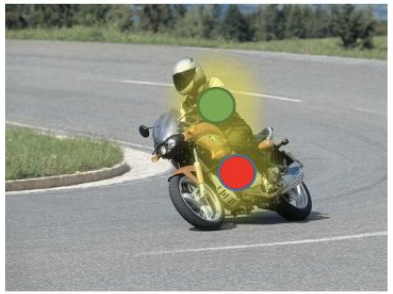
Motivation to Weakly-Supervised Learning

Manual Supervision for Object Recognition

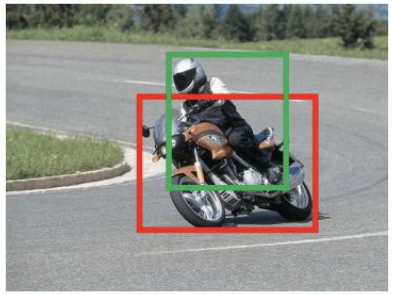
- Weak supervision: Lower degree (or cheaper) annotation at train time than the required output at test time



{motorbike,person}



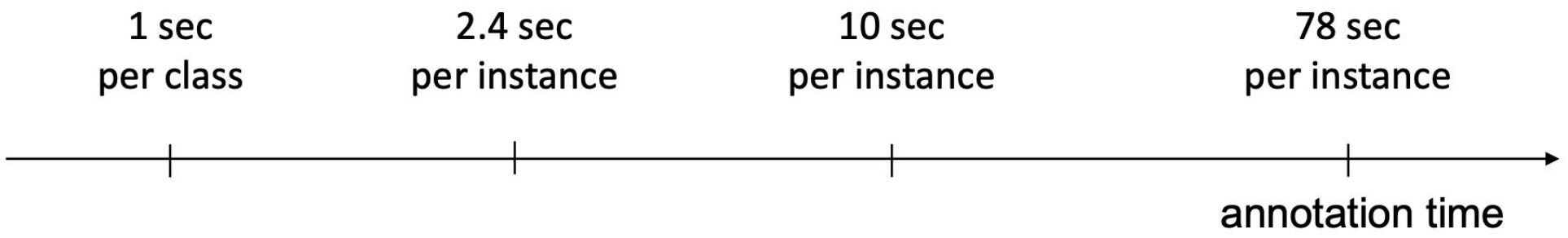
{motorbike (point), person (point)}



{motorbike (b-box), person (b-box)}

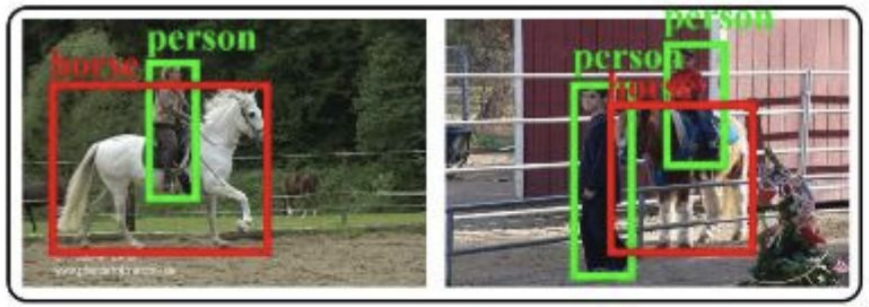


{motorbike (pixel labels), person (pixel labels)}



Berman et al., ECCV 2016, What's the Point: Semantic Segmentation with Point Supervision

Related Topics: Semi-Supervised Learning



(a) Supervised learning



(b) Weakly supervised learning



(c) Weakly semi-supervised learning

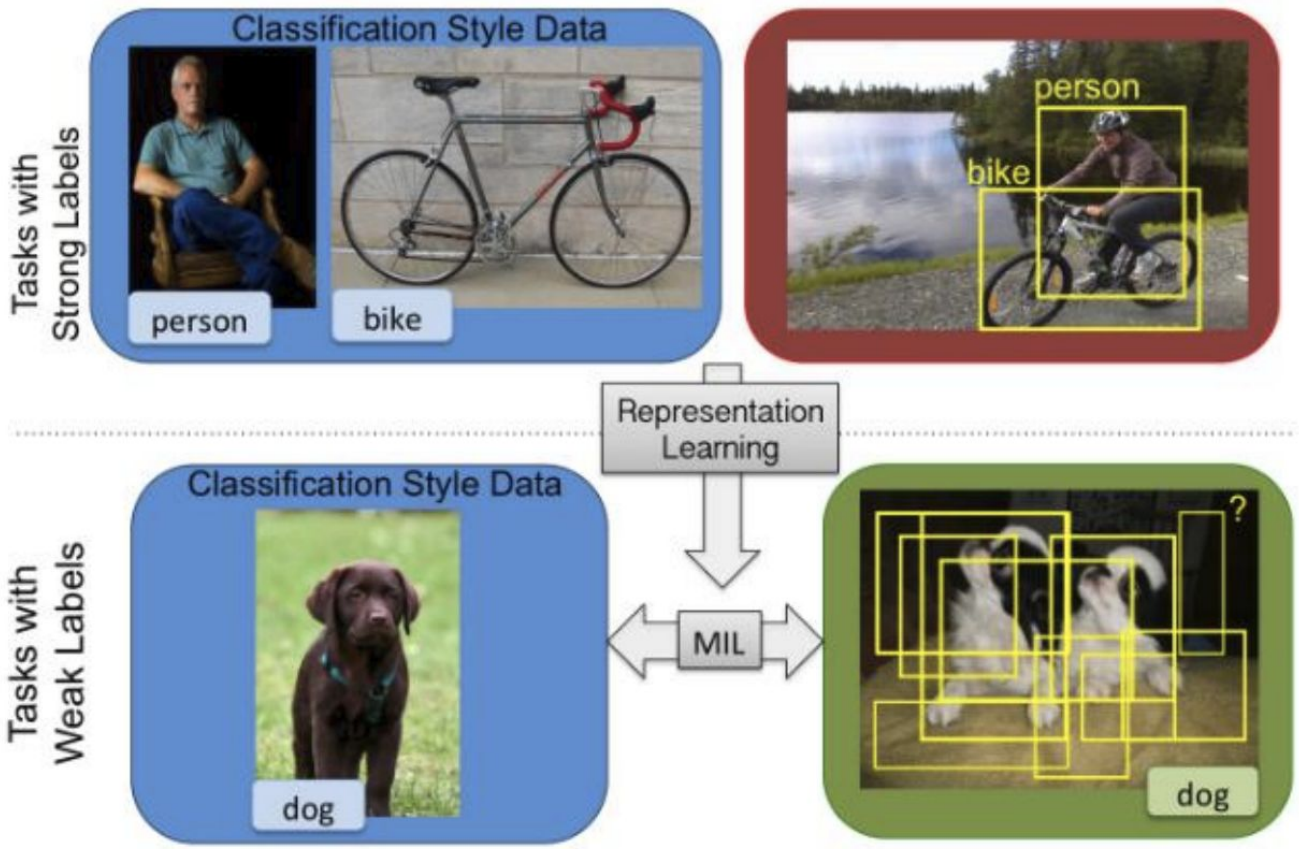


(d) Semi-supervised learning

- **Semi-supervised:** learns from a set of labeled (bounding box) and unlabeled images.
- **Weakly semi-supervised:** learns from a set of labeled (bounding box) and weakly labeled (image label) images.

Image credit: [Jeong 19 NeurIPS]

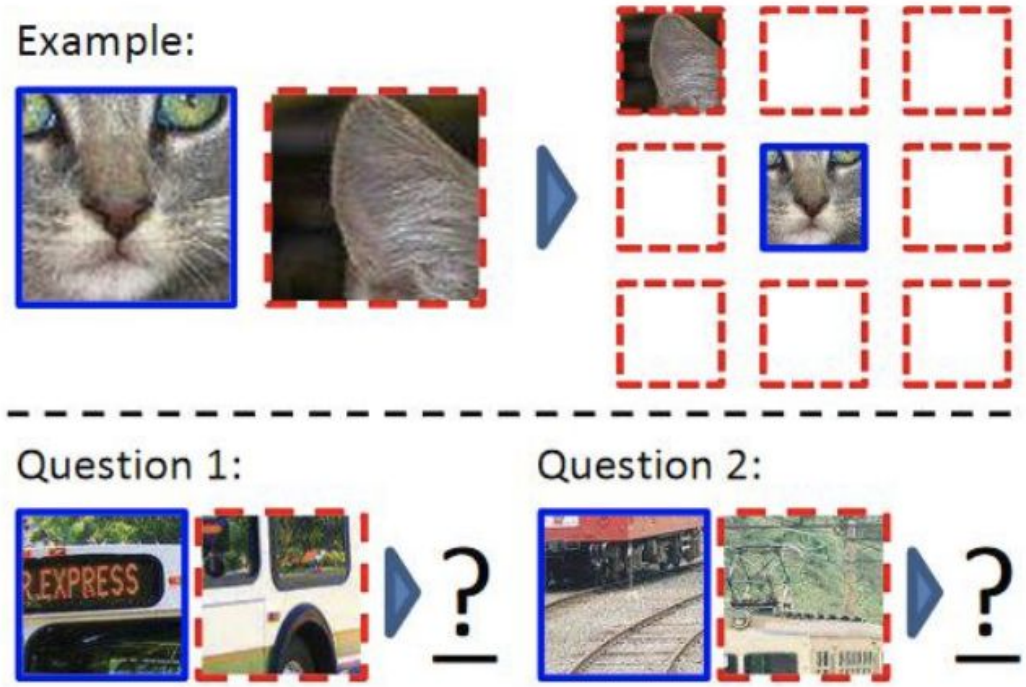
Related Topics: Transfer Learning

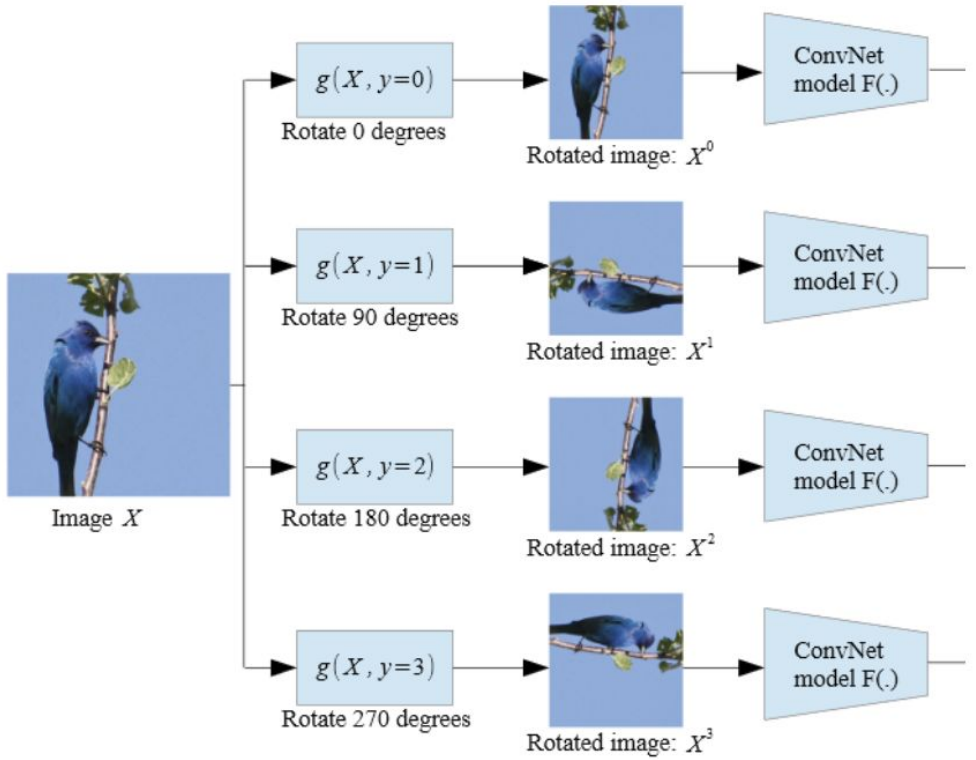


- Learns detectors for auxiliary categories with regular supervision (bounding box)
- Transfers the knowledge from auxiliary categories to weakly supervised categories

Image credit: [Hoffman NIPS 14]

Related Topics: Self-Supervised Learning





- **Defines a pretext task and learns representations in an unsupervised way**
- **Transfers pre-trained weights to a downstream task**
- **Transfer-learning requires regular annotations**

Image credit: [Jeong 19 NeurIPS]

Weakly supervised learning

- Typically uses pretrained networks and fine-tunes them by using weak supervision

Challenges

Challenges in Generic Object Recognition

- Intra-class variations
  - Appearance
  - Viewpoint
  - Scale
  - Aspect ratio
- Background clutter
- Occlusions



Cooccurrence

- Parts vs object



Question: What is a person?
a) Face
b) Face + upper body
c) Face + whole body ☺

Cooccurrence

- Context vs object



Question: What is a train?
a) Rail
b) Train + Rail
c) Train ☺

Cooccurrence

- Multiple vs single instance



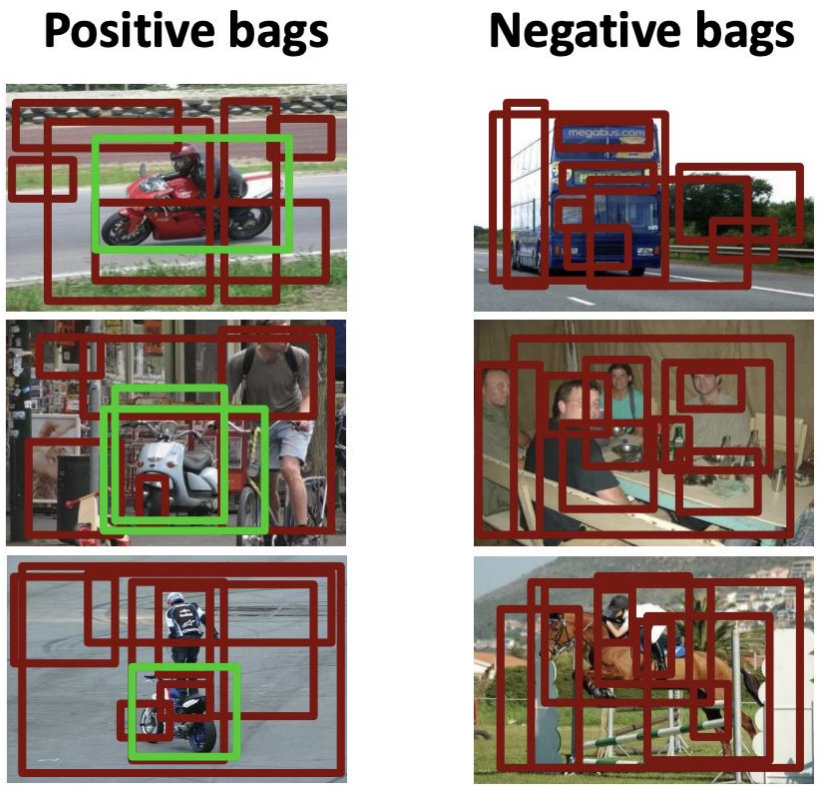



Question: What is a sheep?
a) 2 sheeps
b) 3 sheeps + grass
c) 1 sheep ☺

Weakly-Supervised Object Detection

Principles for WSOD

- Multiple-Instance Learning (MIL)
  - MIL reformulates weakly supervised object detection as a bag classification problem, where each image is a bag of candidate windows, and only the bag label (not the instance labels) is known.



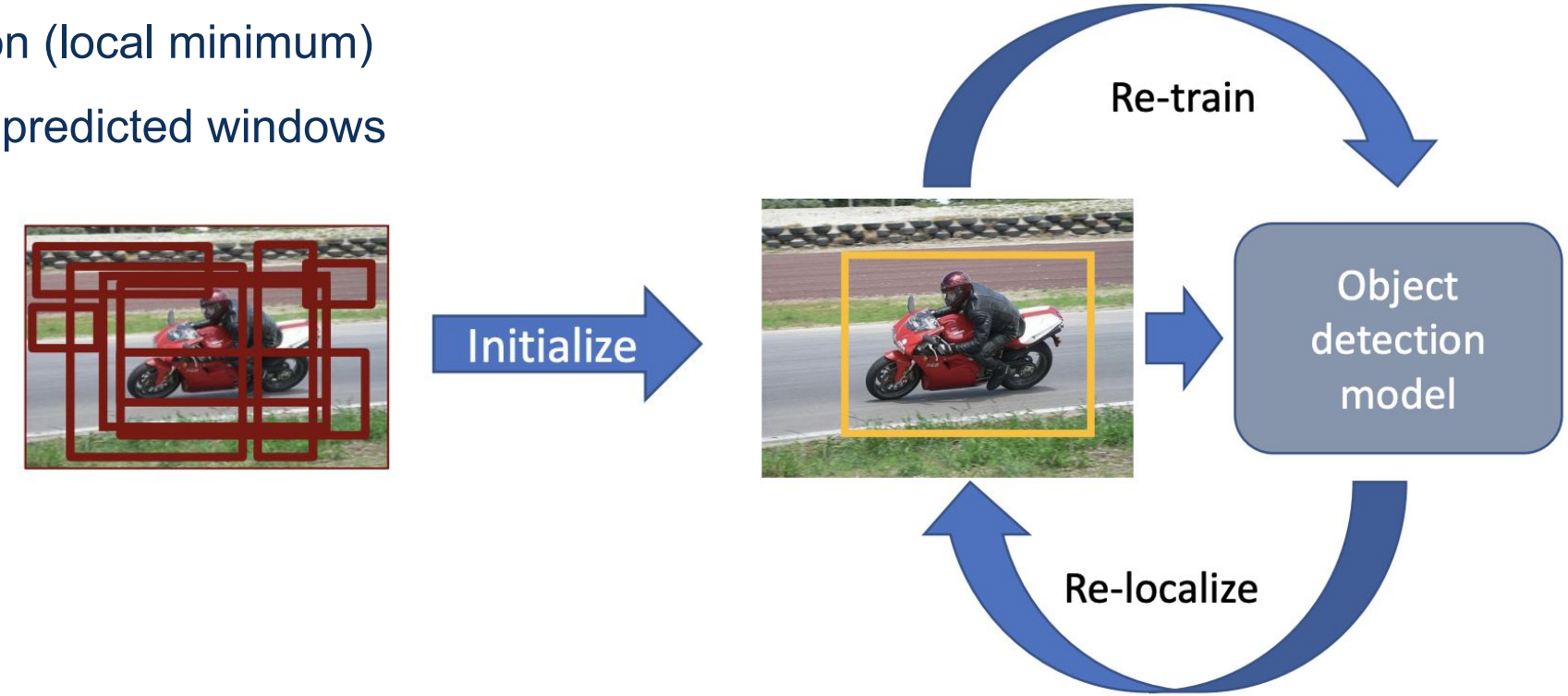
bags = images
instances = windows

Goals:

- find true positive instances
- train window classifier

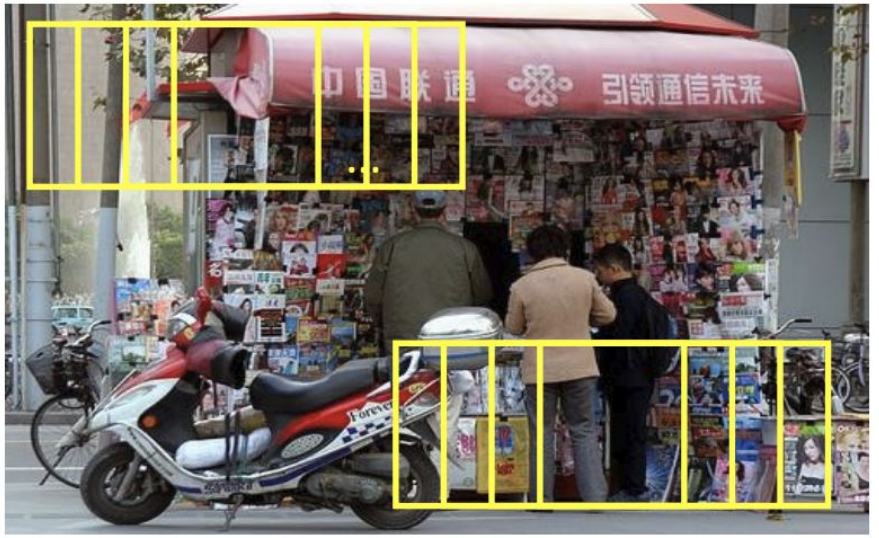
Standard MIL Pipeline

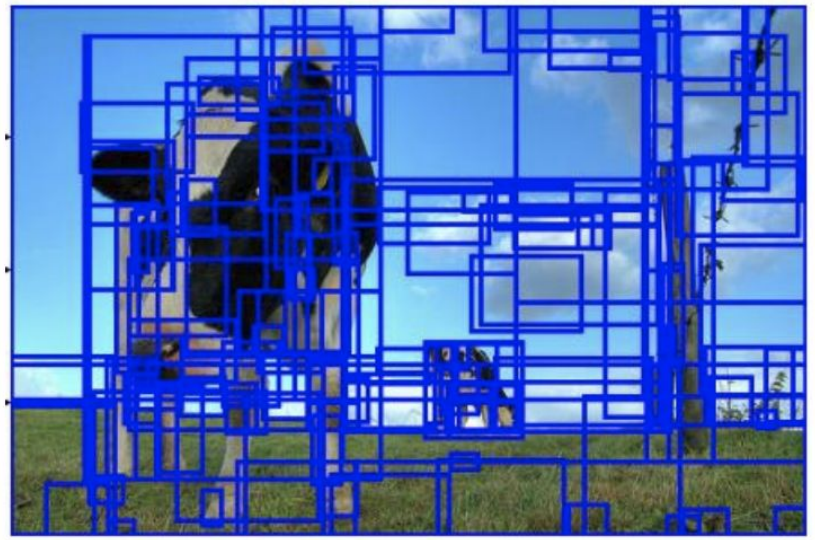
- Two step alternating optimization
  - Re-train
  - Re-localize
- Challenges
  - Sensitive to initialization (local minimum)
  - Overfitting (locking) to predicted windows



How to Generate Bags?

- Sliding windows
  - >100
  - dense
  - translations, scales and aspect-ratios (4D space)
- Object proposals
  - Selective Search and Edge Boxes
  - ~2k per image
  - Commonly used in WSOD





Slide credit: Vittorio Ferrari

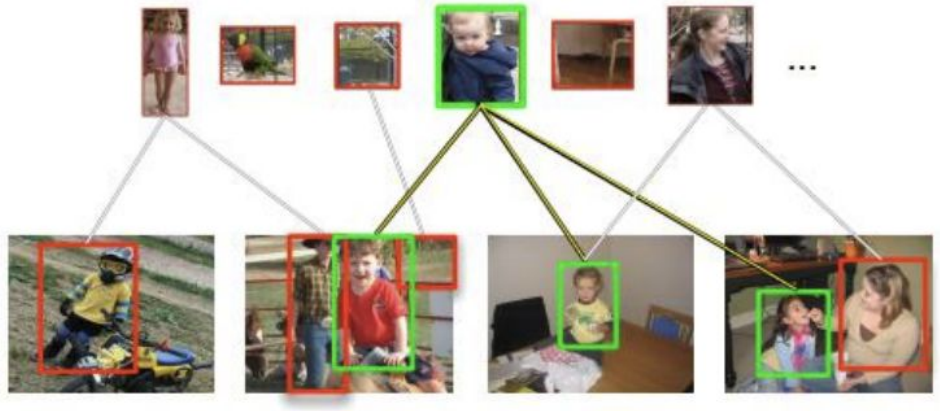
Initialization

- Simple strategies:
  - Whole image
  - Whole image minus a margin
- [Song ICML 14] constructs a graph to find initial boxes:
  - 1. relevant (occur in many positive images)
  - 2. discriminative (dissimilar to the boxes in the negative images)
  - 3. complementary (capture multiple modes)

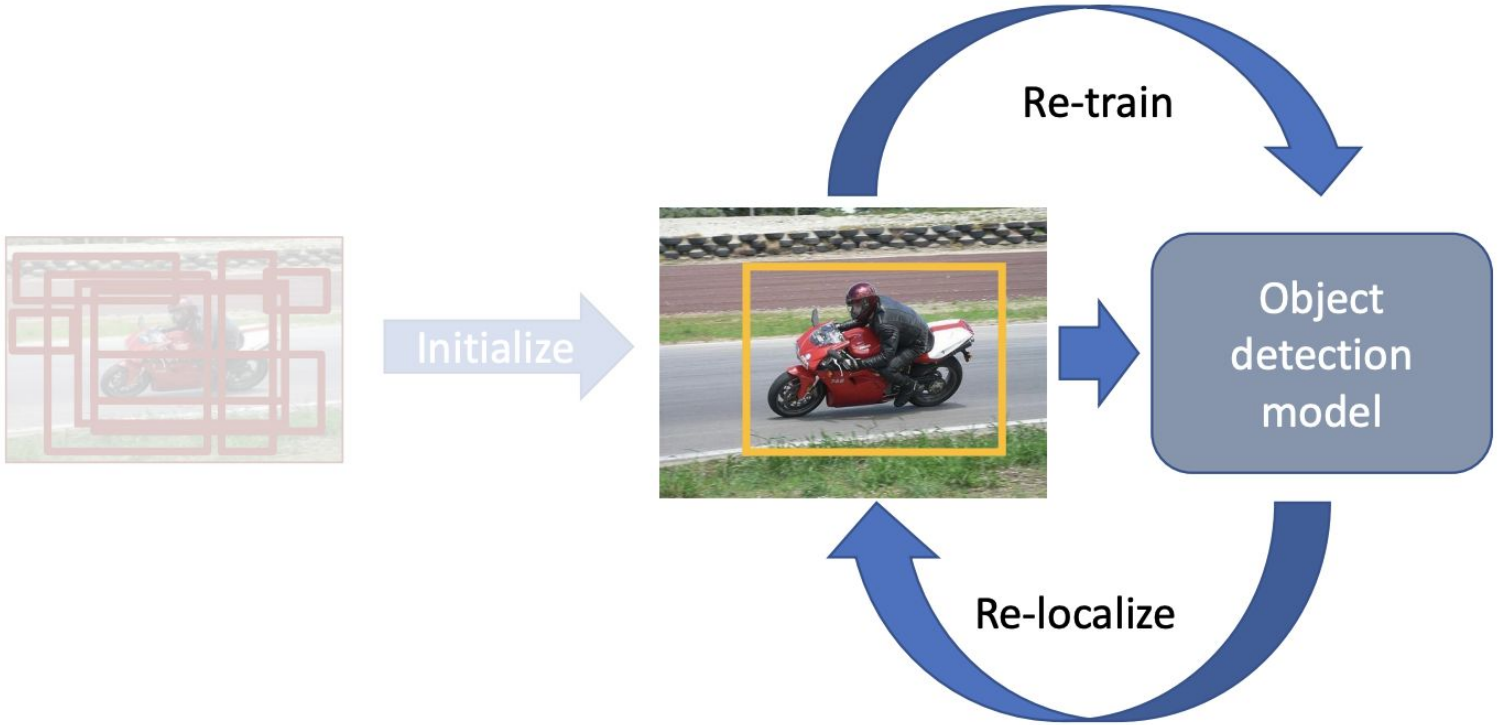
Instead of initializing MIL with the entire image, Song et al. construct a proposal graph across positive images and select object proposals that are (1) common across positives, (2) discriminative from negatives, and (3) representative of diverse object appearances.





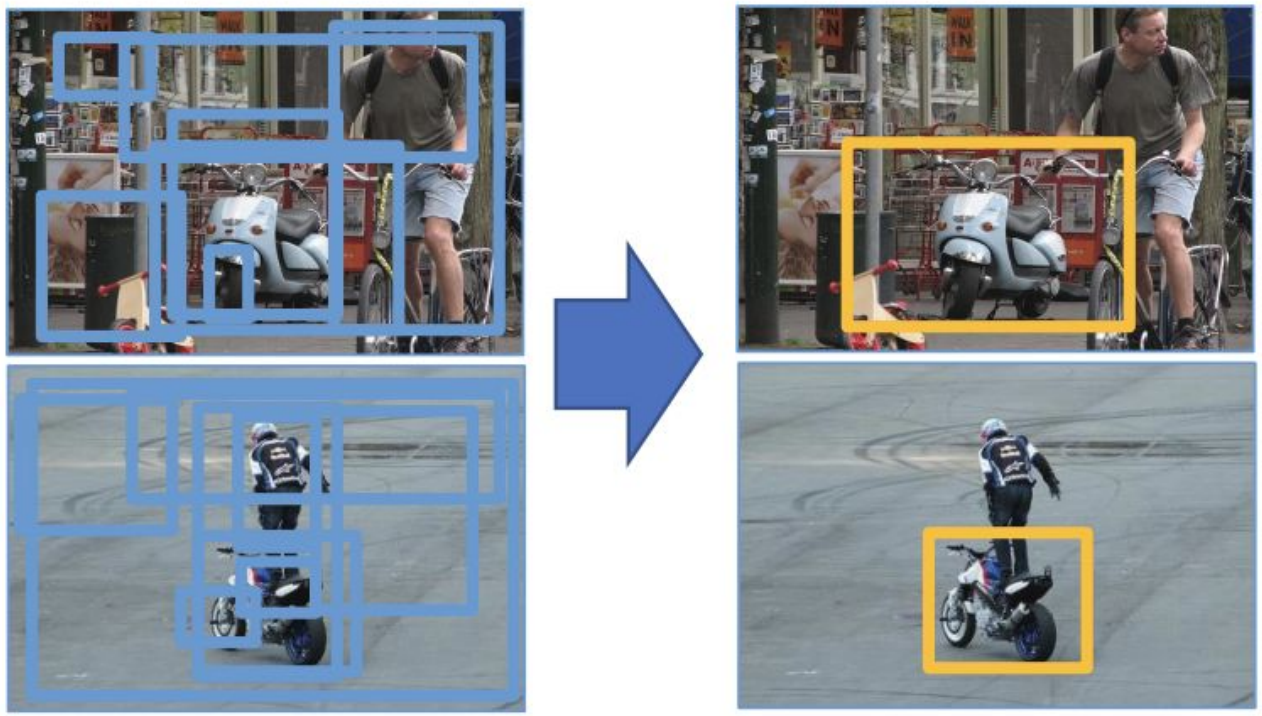


Standard MIL Pipeline



Re-localize

- Standard max formulation
  - Pick the highest scoring window as positive
  - Only one positive instance per image



$$\arg\max_{b} A(x_b)$$

Proposal — Appearance model — Features

[Nguyen ICCV 09, Bilen CVPR 14, Cinbis CVPR 14, Papadopoulos CVPR 16]

Re-train

- Hinge and cross-entropy loss

For positive images:

$$\max_b A(x_b) > \Delta \text{ (}\Delta\text{:margin) or } \log(\text{Softmax}(\max_b A(x_b)))$$

For negative images:

$$\max_b A(x_b) < -\Delta \quad \text{or} \quad \log(1-\text{Softmax}(\max_b A(x_b)))$$

Difference to supervised object detection (eg Faster-RCNN)

- Only 1 positive instance is used in each positive image
- No negative instances from positive image

During the re-training stage, MIL selects the highest-scoring proposal as the positive instance for each positive image, while all proposals in negative images are treated as negatives. Unlike fully supervised detection, only one positive proposal is used and no negative proposals are sampled from positive images.

[Nguyen ICCV 09, Bilen CVPR 14, Cinbis CVPR 14, Papadopoulos CVPR 16]

Re-Localization and Re-Training

- More robust optimization: Relaxing max operator
- Hedge your bets on multiple proposals

Re-train

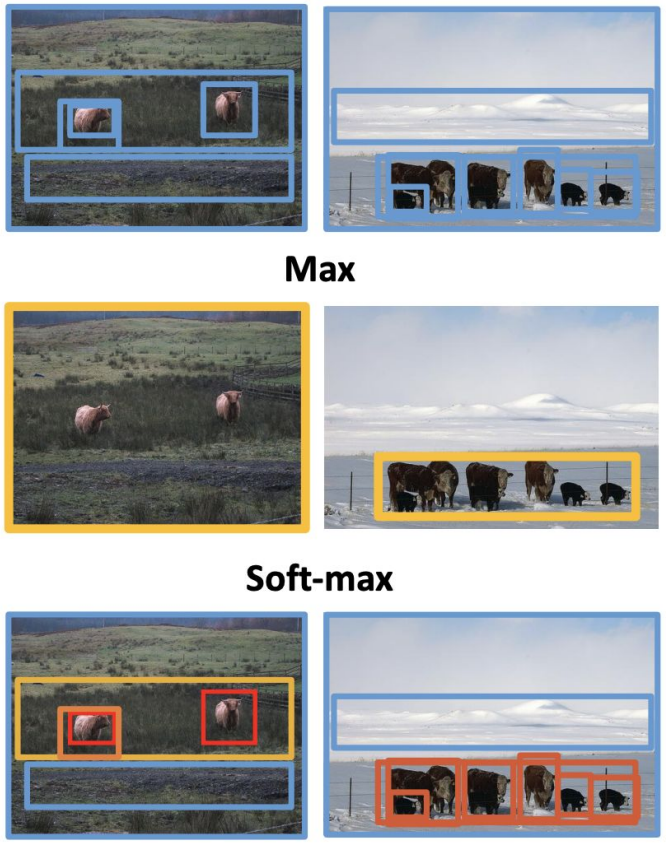
For positive images

$$\log \sum_b \exp A(x_b^+) > \Delta$$

For negative images

$$\log \sum_b \exp A(x_b^-) < -\Delta$$

- Less sensitive to initialization
- Leverages multiple positive samples during training



Relax hard max into soft-max (log-sum-exp) to learn from multiple high-scoring proposals instead of a single proposal.

[Bilen BMVC 14]

Re-Localization and Re-Training

- More robust optimization: Self-paced learning
- In conventional WSOD, the initial localization results are crucial.
- If an incorrect bounding box is selected initially,
  - → the detector is trained using that box,
  - → leading to a vicious cycle (locking) where that same box is repeatedly selected.
- Idea: Inspired from Curriculum Learning, Instead of training on all samples—including difficult ones—from the start, let's train progressively, beginning with easy samples.
- Easiness measures for WSOD:
  - Selection of samples via confidence of max scoring window
  - Selection of window space by allowing smaller windows
  - Selection of samples via intercategory competition



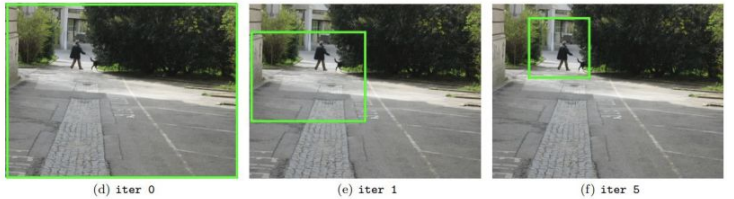
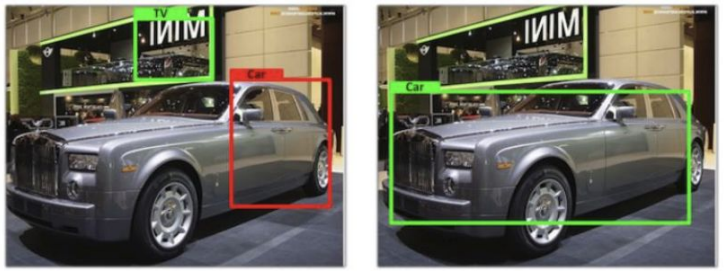


Image credit: [Bilen IJCV 14]

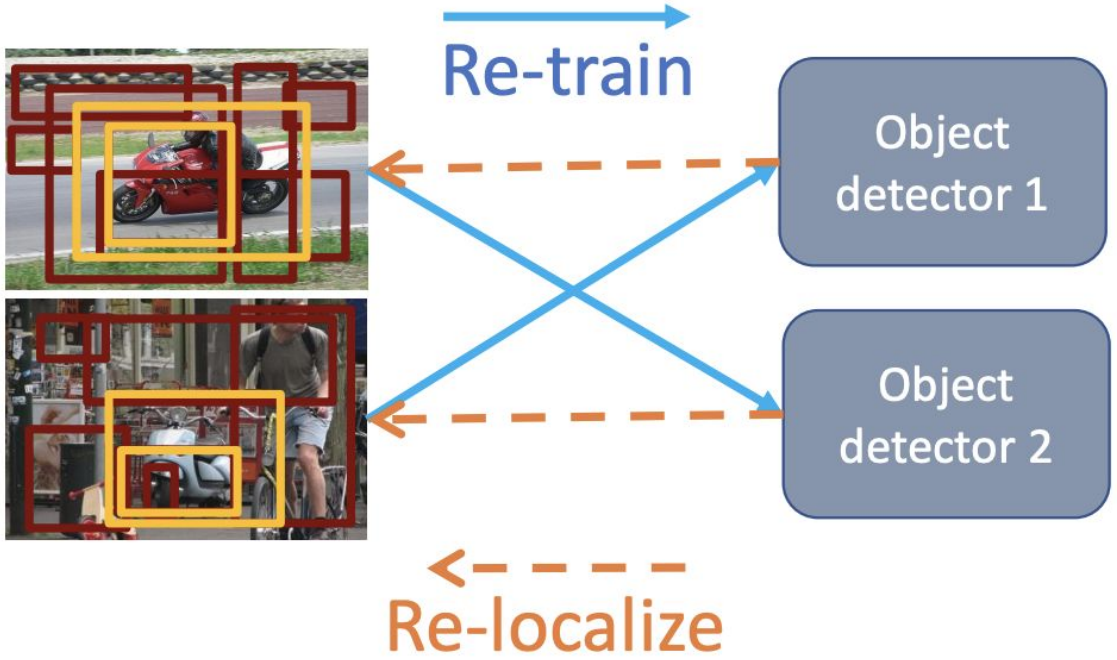


Learn from easy images first, then progressively incorporate harder samples as localization becomes more reliable.

Image credit: [Sangineto PAMI 17]

Re-Localization and Re-Training

- More robust re-localization: Multifold MIL
  - Problem: Detector overfits into the given proposal
  - Solution: Train using positive examples in all folds but k, and all negative examples



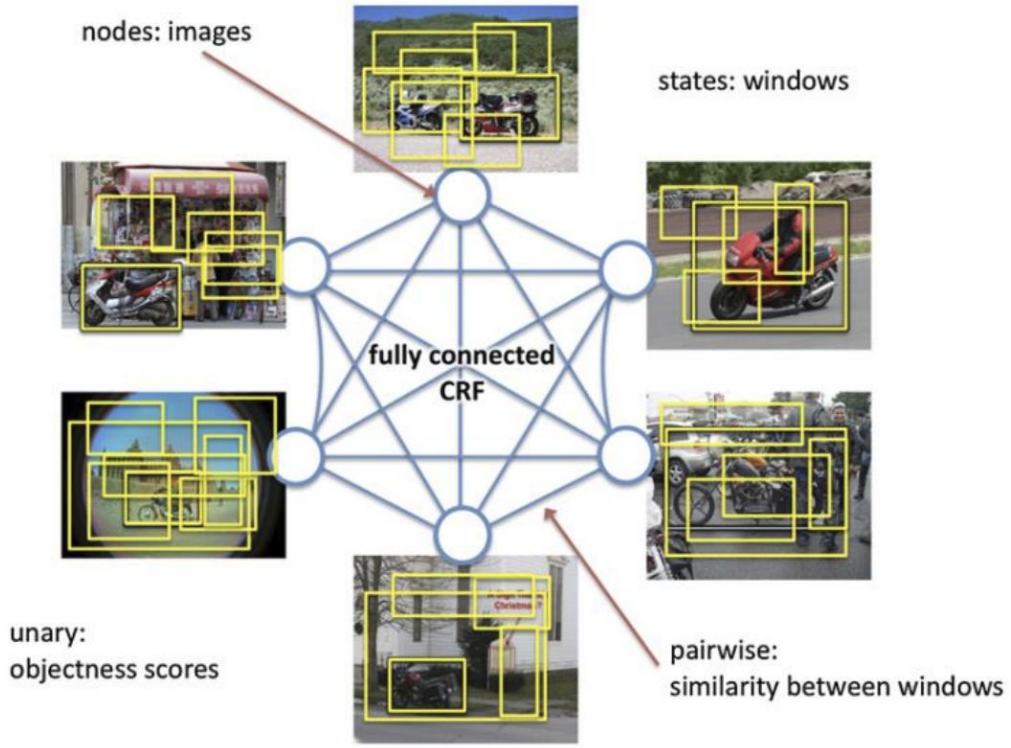
Idea: the detector performing localization for an image must be one that has not been trained on that image.

-> **Use cross-validation style training: localize each image with a detector that has never seen that image during training.**

Image credit: [Sangineto PAMI 17]

Re-Localization and Re-Training

- Pairwise similarity
  - Similarity between selected windows across positive images
    - Less overfitting
    - Expensive to optimize
      - All positive images must be connected (N^2 similarities)
    - Ignores intra-class variation
      - The assumption that all objects must be similar is unrealistic.
      - However, since this method assumes that similar = good, it does not sufficiently reflect diversity.

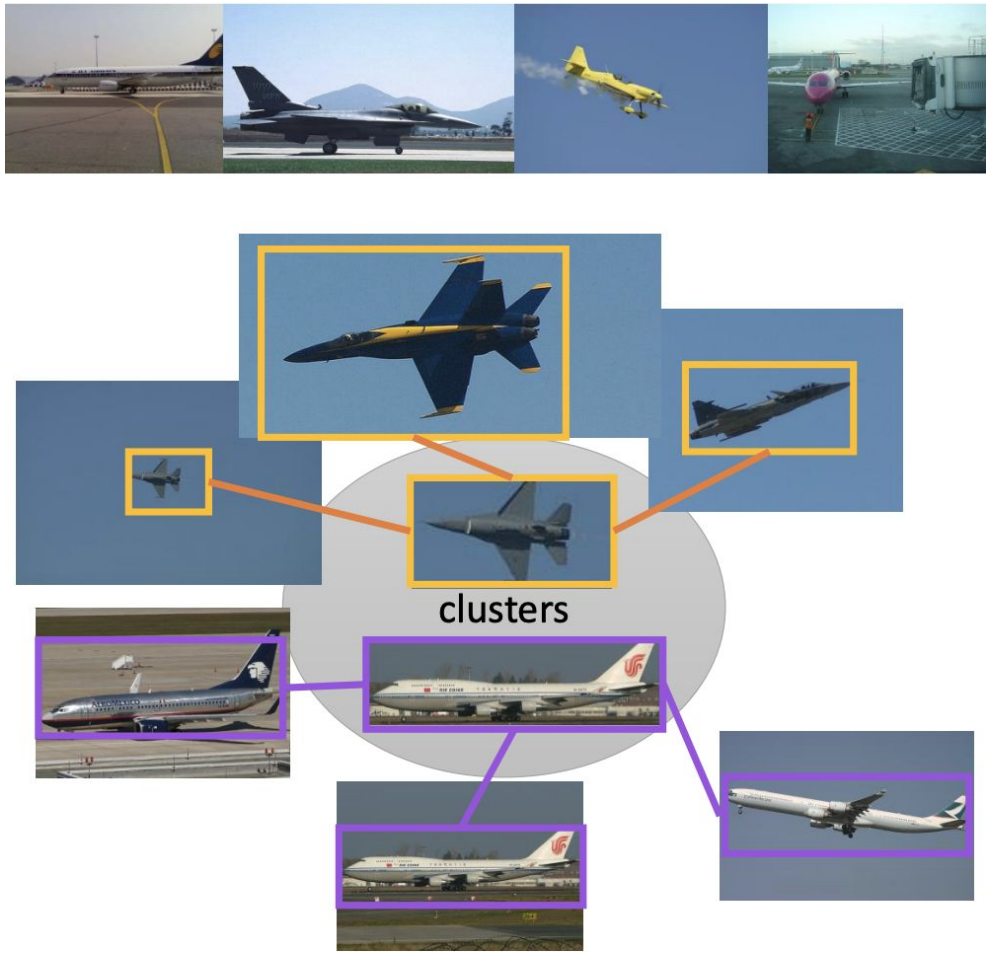


Enforce consistency across positive images by jointly selecting similar object proposals, reducing localization ambiguity but increasing computational cost and limiting intra-class diversity.

[Chum CVPR 07, Deselaers ECCV 10, Siva ICCV 11, Bilen CVPR 15]

Re-Localization and Re-Training

- Pairwise similarity
  - Sub-categories
    - Clustering via probabilistic latent Semantic Analysis (pLSA)
      - (+) Modeling intra-class variations
      - (-) Sensitive to number of clusters
  - Exemplars
    - Pick representative samples
    - (+) No need to set number of clusters

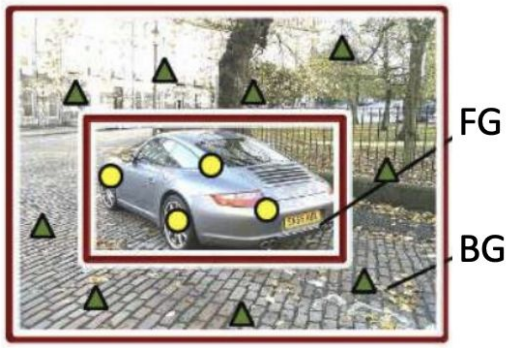


Instead of enforcing similarity across all positive images, model intra-class variation using either sub-category clustering or representative exemplars.

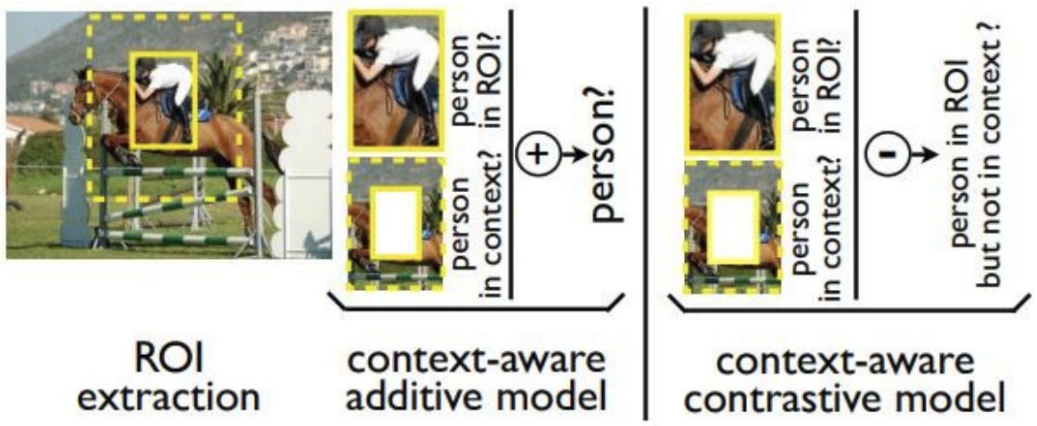
[Wang ECCV 14, Chum CVPR 07, Bilen CVPR 15]

Re-Localization and Re-Training

- Additional cues: context
  - Background provides contextual cues for recognition
  - Better separation of foreground and background
  - Additive: select a ROI that is semantically compatible with its context
  - Contrastive: select a ROI that is outstanding from its context



[Russakovsky ECCV 12]



[Kantorov ECCV 16]

Context provides valuable cues beyond the object appearance. It can either reinforce semantically compatible objects (additive) or help distinguish foreground from background (contrastive).

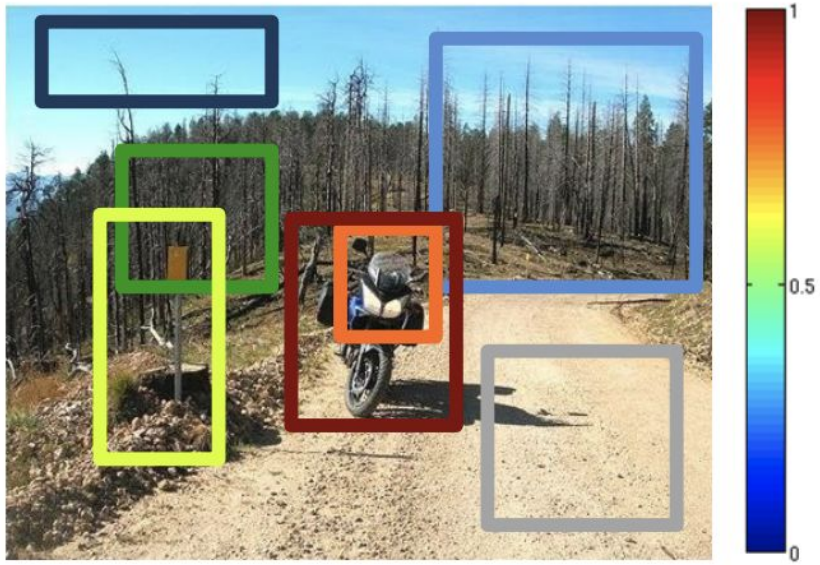
Re-Localization and Re-Training

- Additional cues: Objectness
  - Quantify how likely a window is to contain an object of any class
    - Generates a small set of accurate object proposals
    - Steers re-localization towards objects and away from background
    - Pushes towards whole objects instead of sub-regions

$$argmax_b \lambda A(x_b) + (1-\lambda)\mathrm{Obj}(b)$$

Obj(b): proposal의 objectness
- EdgeBoxes
- Selective Search
- BING

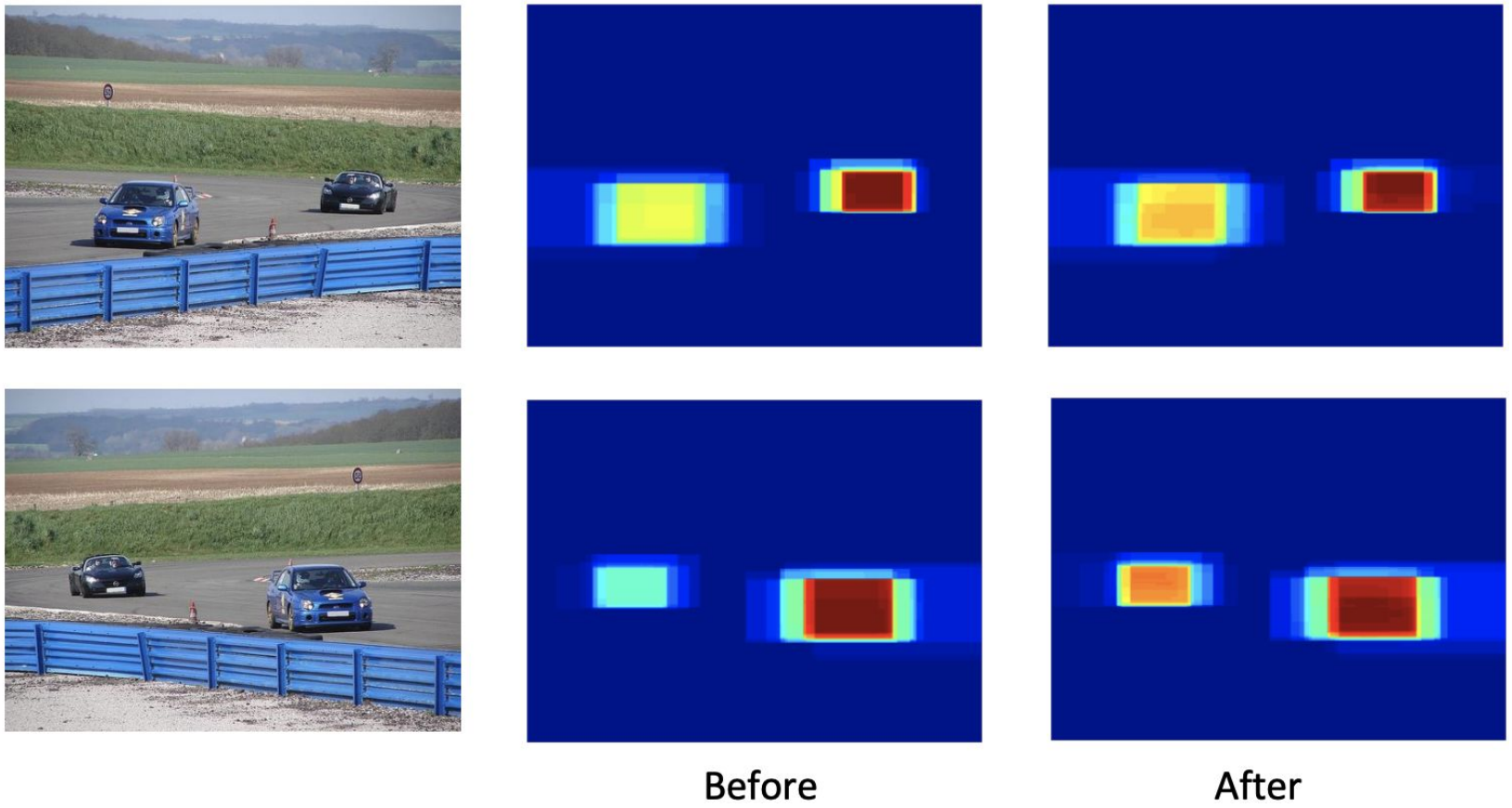
Objectness provides a class-independent prior that guides WSOD toward complete objects rather than discriminative object parts or background regions.



Slide credit: Vittorio Ferrari

Re-Localization and Re-Training

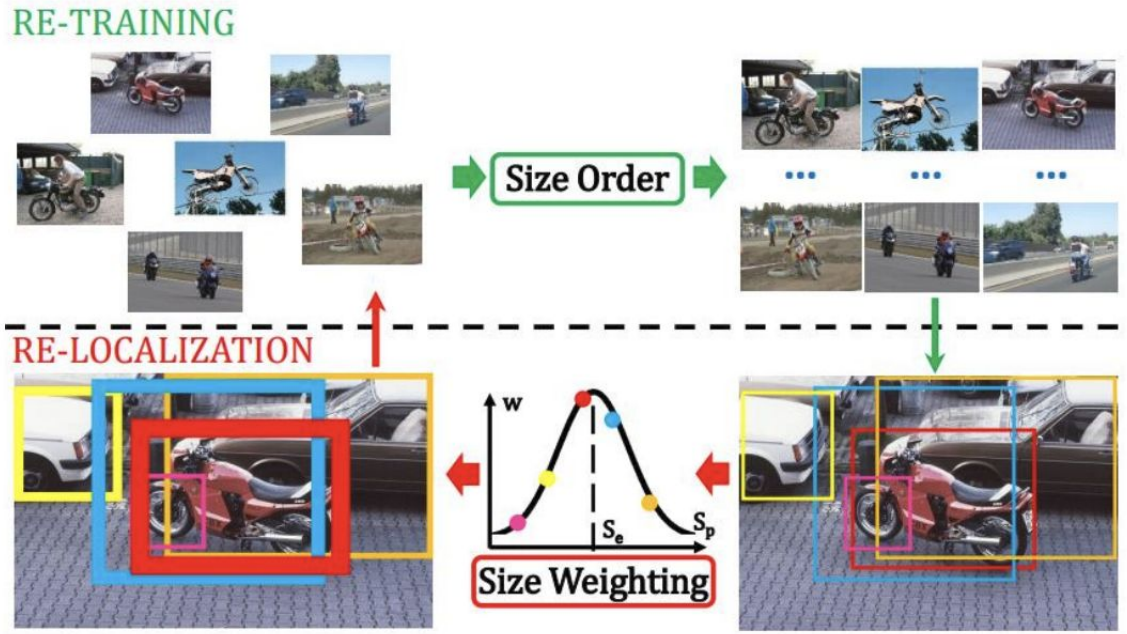
- Priors: Equivariance
  - The predicted output should be equivariant to image transformations



[Bilen BMVC 14]

Re-Localization and Re-Training

- Priors: Scale
  - Curriculum learning (bigger objects down to smaller ones)
  - Weight object proposals according to estimated size
  - Requires training a size estimator from a small set



introduces an object-size prior, encouraging the detector to localize large objects first and progressively learn smaller ones, thereby reducing localization errors caused by object parts.

[Shi ECCV 16]

Re-Localization and Re-Training

- Priors: Motion
  - 1. extracts candidate spatio-temporal tubes based on motion segmentation
  - 2. selects one tube per video jointly over all videos
  - 3. uses objectness to reduce object candidates
  - 4. applies a domain adaptation method to address the video/image gap
  - (+) Motion cues for object boundaries
  - (-) Noisy data

Motion provides strong temporal cues for object localization. By extracting motion-based object tubes from videos and transferring them to image detectors, WSOD can obtain much more reliable pseudo bounding boxes than using images alone.

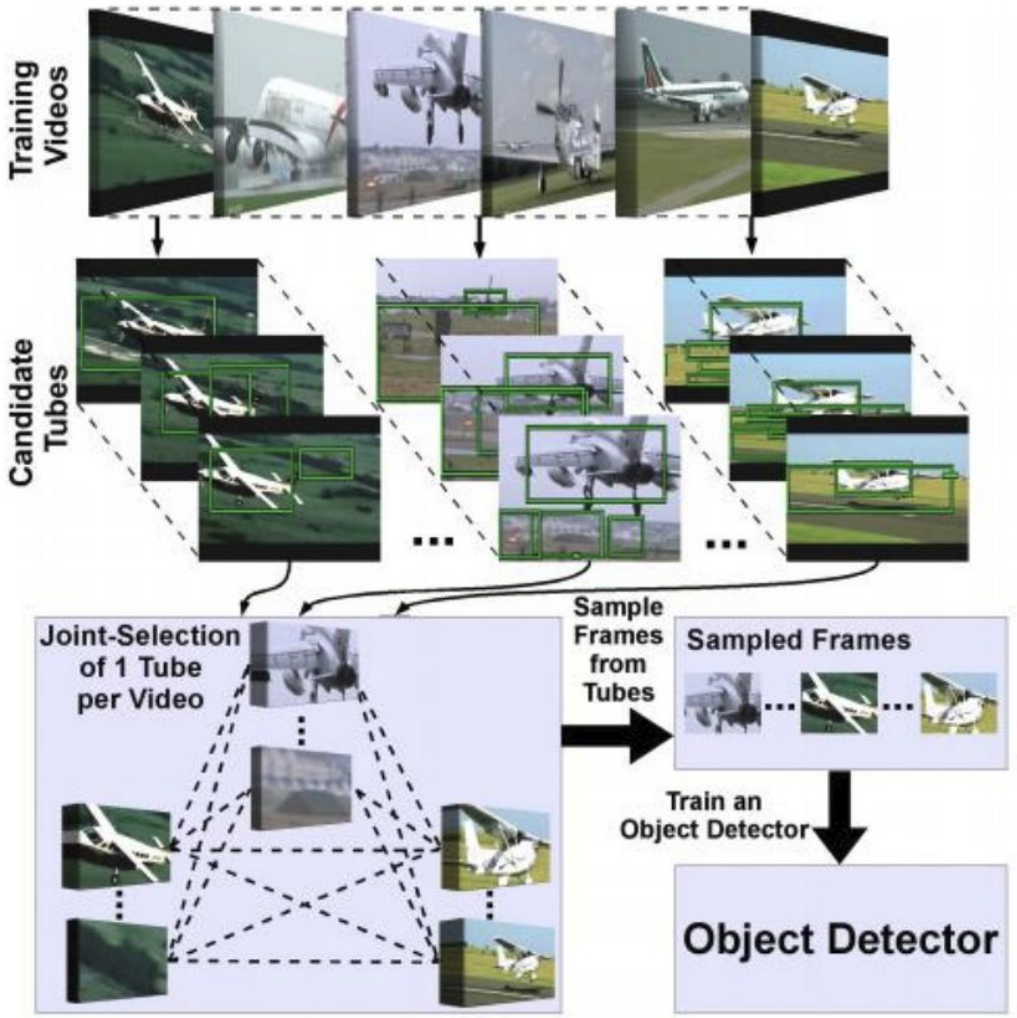
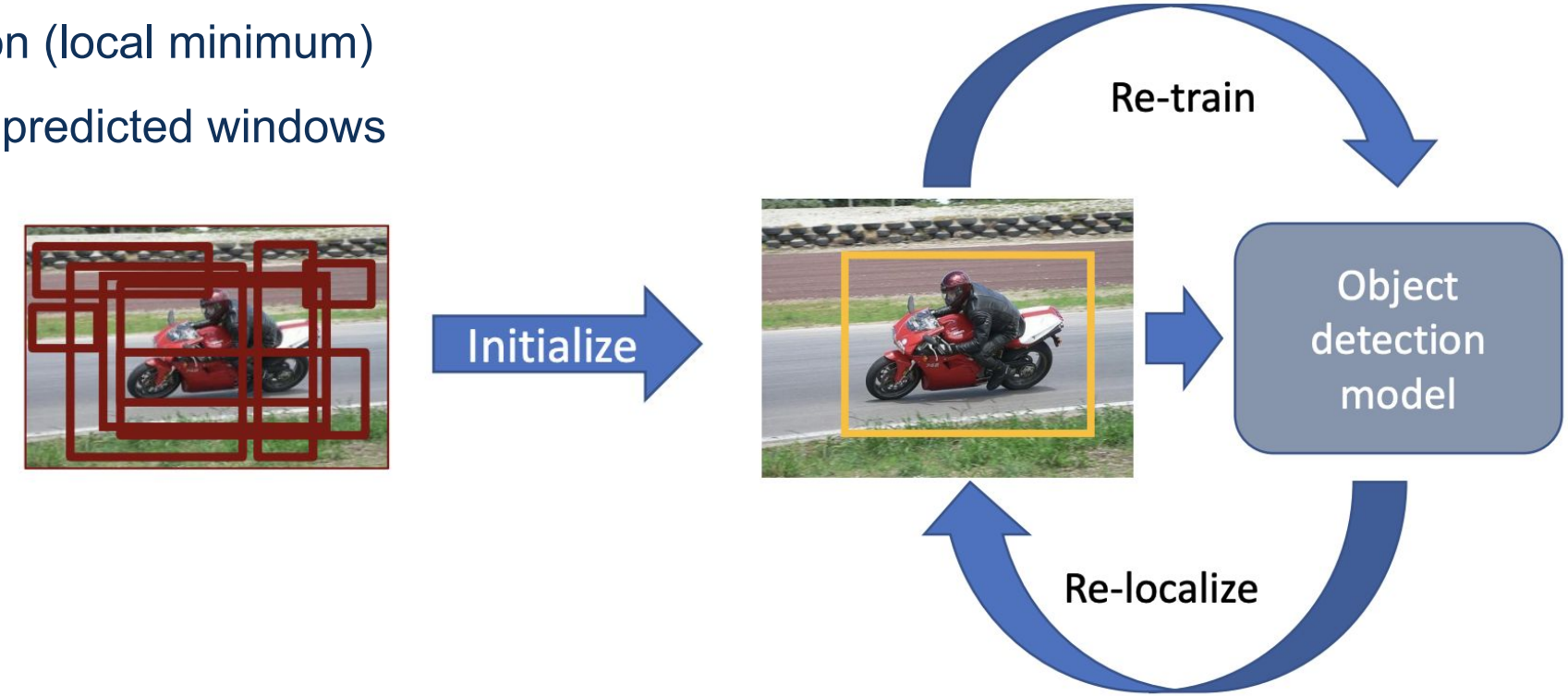


Image credit: [Prest CVPR 12]

Standard MIL Pipeline

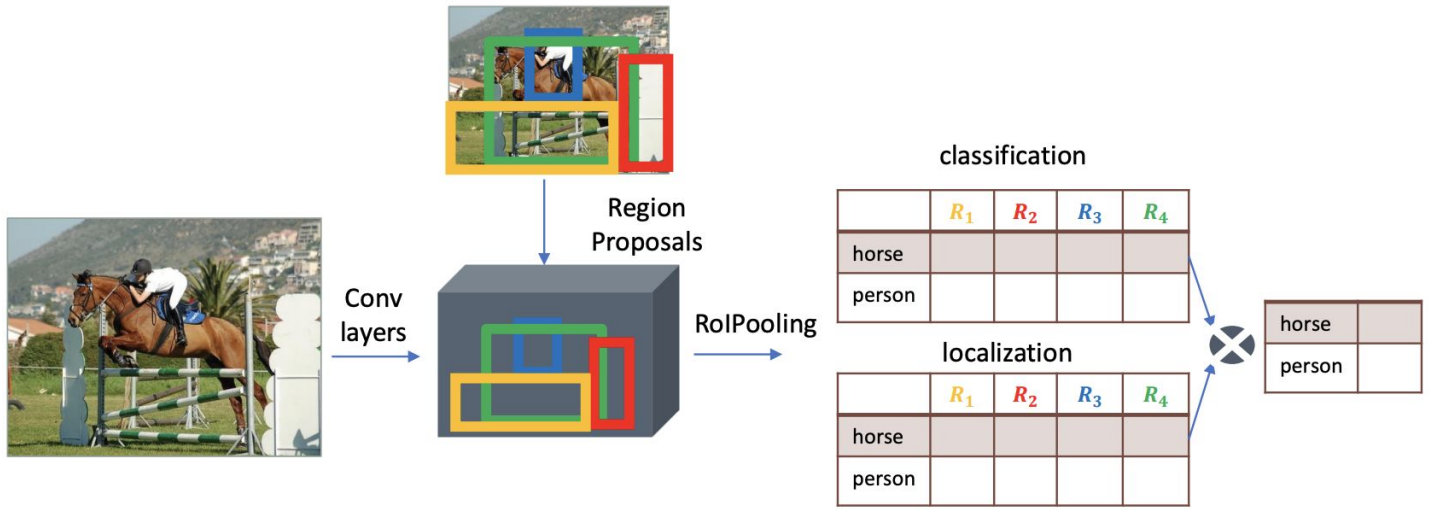
- Two step alternating optimization
  - Re-train
  - Re-localize
- Challenges
  - Sensitive to initialization (local minimum)
  - Overfitting (locking) to predicted windows



Deep and End-to-End WSOD

Weakly Supervised Deep Detection Network (WSDDN)

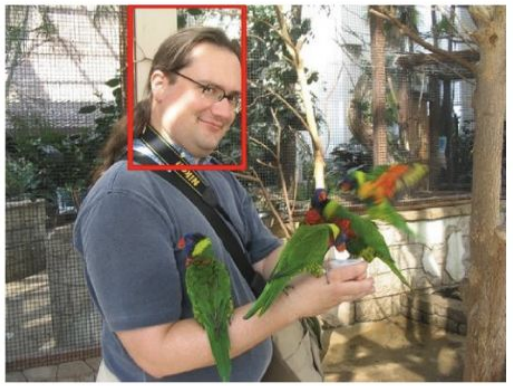
- Previous work uses CNN as black-box feature extractor
- First end-to-end WSOD method, finetunes a pre-trained deep network
- Uses RoI-Pooling
- Two stream architecture
  - Classification: Associates each region with a class score
  - Localization: Compares regions for each class

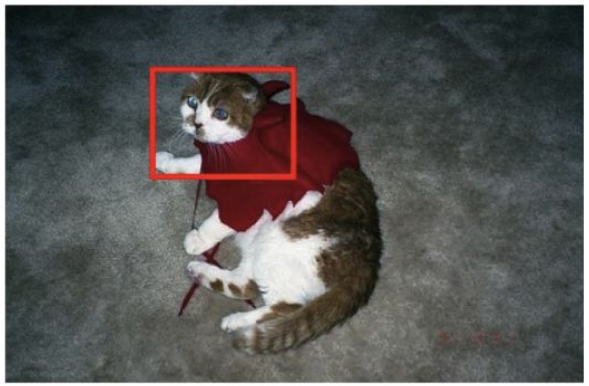


[Bilen CVPR 16]

Weakly Supervised Deep Detection Network (WSDDN)

- (+) State-of-the-art results with VGG-F (~AlexNet) (62% of supervised)
- (+) End-to-end learning + No custom deep learning layers
- (-) Detects discriminative parts rather than whole object extent
- (-) Groups multiple instances together
- Limitations (compared to the supervised case, eg Faster RCNN):
  - It does not leverage multiple positive instances in positive images
  - It does not exploit negative instances in positive images
  - It does not learn bounding box regression



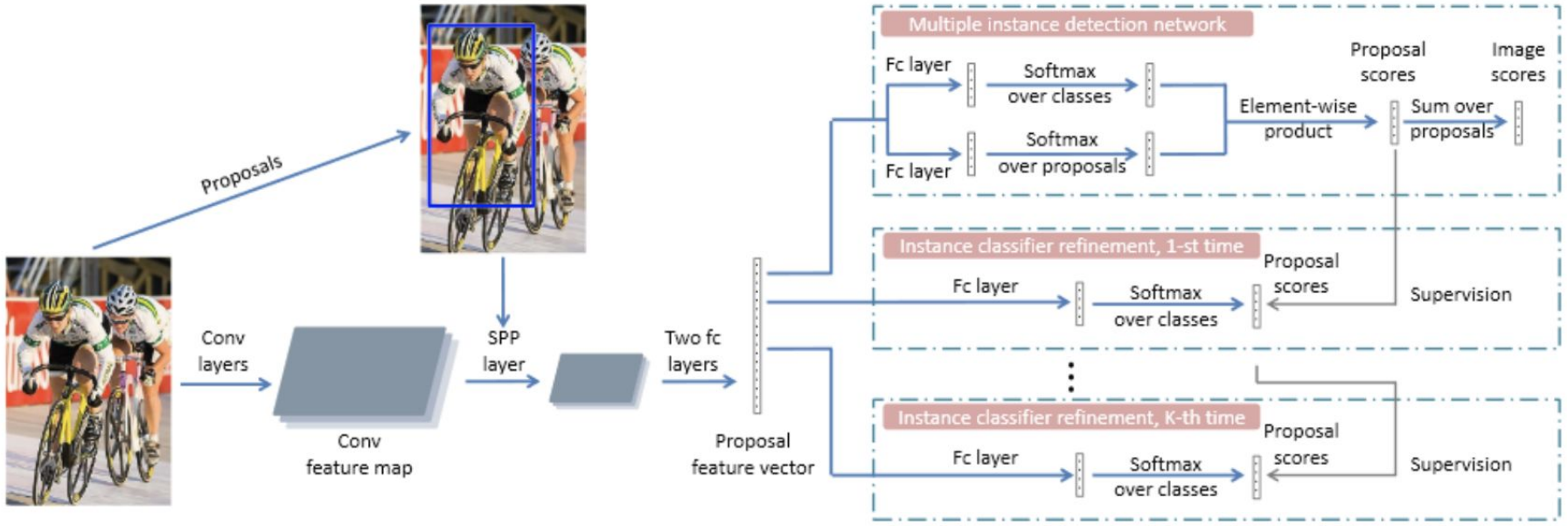




[Bilen CVPR 16]

Incorporating Instance Classification

- **WSDDN** learns to classify images based on implicit instance scoring
- Online Instance Classifier Refinement (**OICR**)
- **OICR** learns to classify instances
  - Initially uses WSDDN to set instance labels
  - Iteratively refines them by using an instance classifier

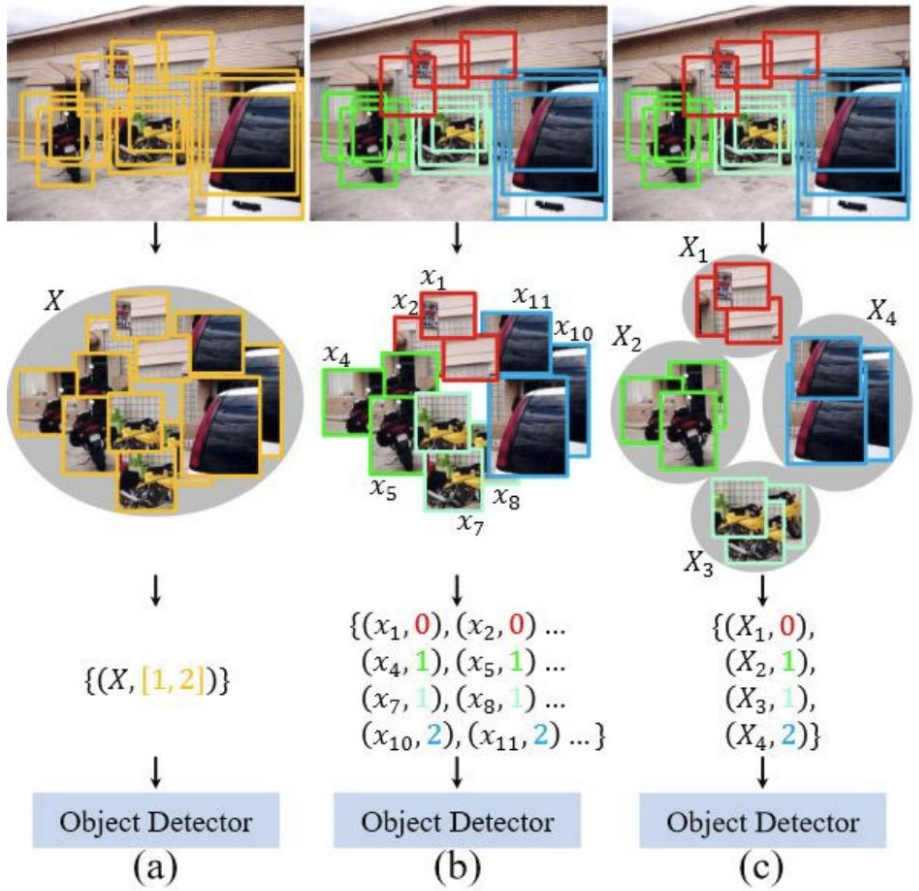


WSDDN -> Pseudo labels -> Instance classifier -> Better pseudo labels -> Better instance classifier --> .. ç

[Tang CVPR 17]

Incorporating Instance Classification

- Proposal Cluster Learning (PCL) for WSOD
  - **Observation**: There can be multiple positive instances in an image
  - **OICR** considers only the highest scoring proposal + surrounding ones as positives
  - **PCL**
    - builds an undirected graph of top- ranking proposals
    - iteratively selects vertices which have most connections to be the cluster centres



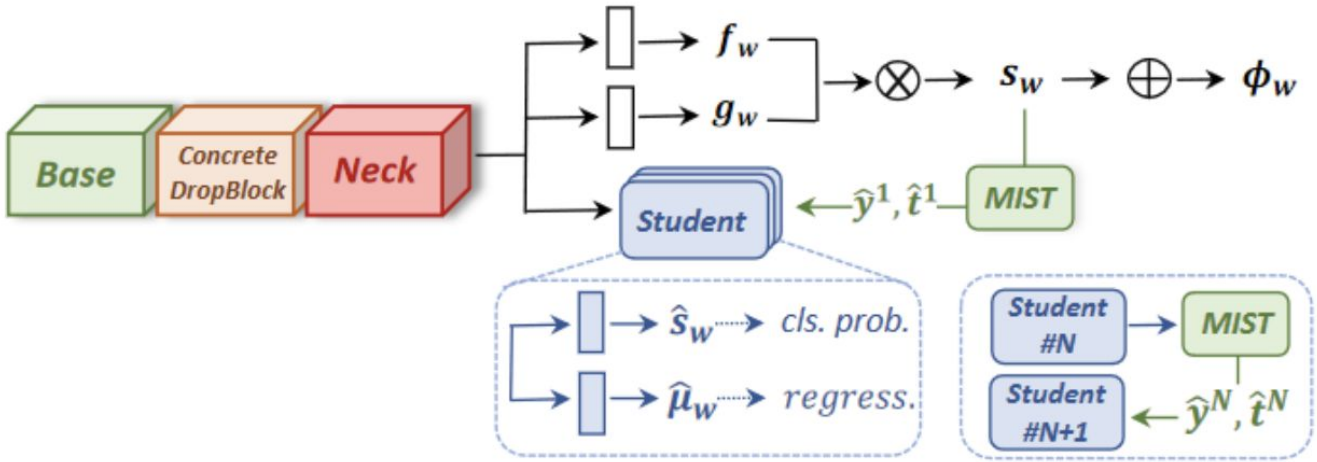
OICR uses a single pseudo object per image, whereas PCL recognizes that an image may contain multiple object instances. It groups high-scoring proposals into clusters and uses each cluster as an independent pseudo object, leading to more accurate instance-level supervision.

[Tang PAMI 18]

Regularizing Discriminative Parts

- Instance-Aware, Context-Focused, and Memory-Efficient WSOD
  - Builds on WSDDN and OICR
    - Picks top p % of the highest scoring proposals and surrounding ones as positives
    - Introduces ConcreteDropBlock: a data-driven and parametric drop-out to drop the most relevant regions
    - Adversarial optimization

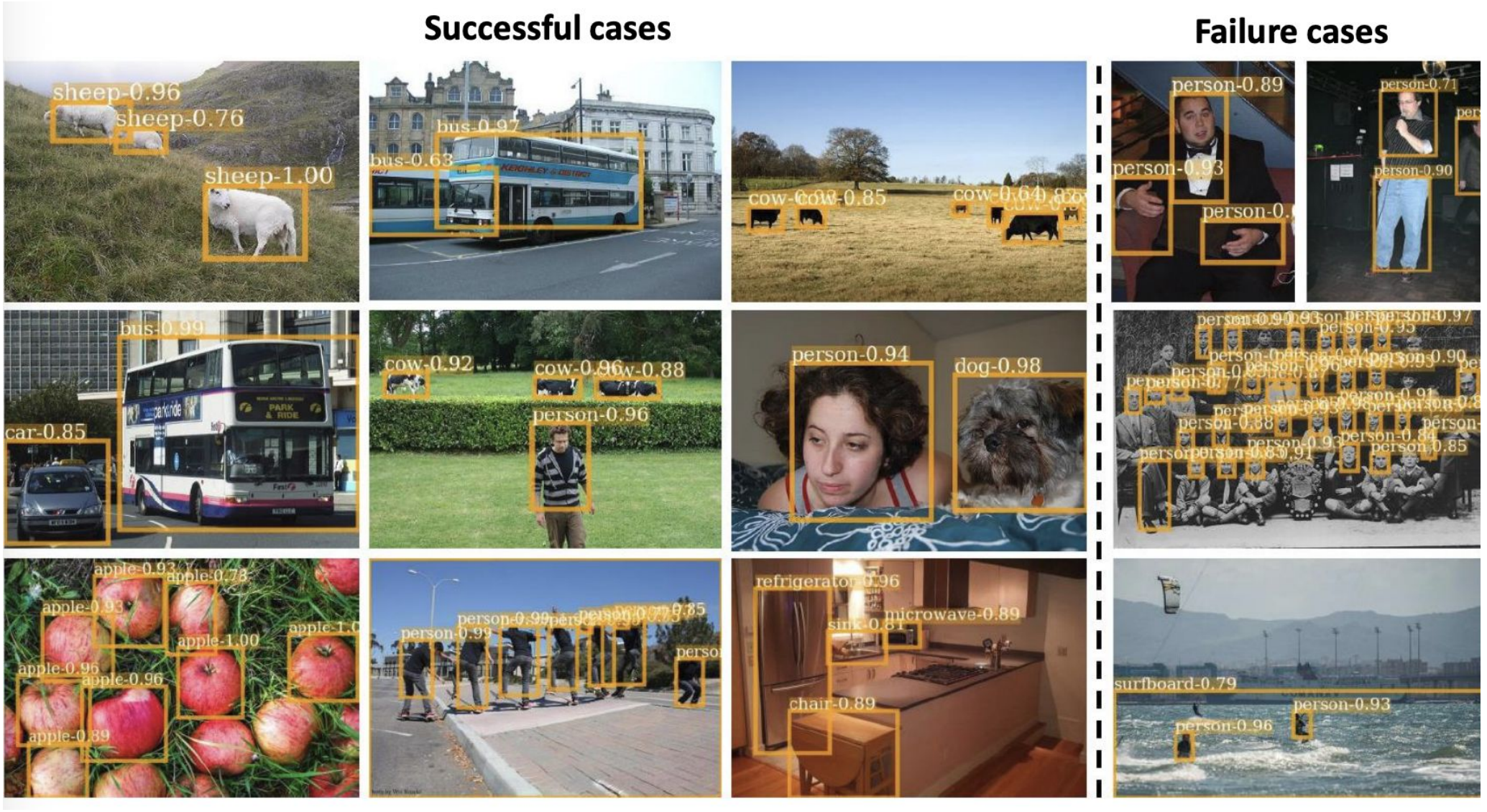
$$w^*, \theta^* = \arg\min_{w} \max_{\theta} \sum_{I} \mathcal{L}_{\text{img}}(w, \theta) + \mathcal{L}_{\text{roi}}(w, \theta)$$



Unlike OICR and PCL that mainly improve pseudo-label quality, this work explicitly addresses the discriminative-part problem. By suppressing the most discriminative regions using Concrete DropBlock, the detector is forced to attend to the entire object rather than only its most salient parts, leading to better localization.

[Ren CVPR 20]

Regularizing Discriminative Parts



Summary on Weakly-Supervised Object Detection

- WSOD is an ill-posed problem and ambiguous due to part/whole and co-occurence issues.
- WSOD training is sensitive to initialization and prone to overfitting.
- Remarkable progress due to
  - objectness and region proposals,
  - incorporating prior knowledge and additional cues,
  - end-to-end learning.
- Addressing the inherent ambiguity requires better objectness measures and stronger additional cues.

Next Week..

- Weakly supervised ..